

Distribution-free comparison of fasting glucose by group

Abstract

Background: A dietary intervention study on fasting glucose. Methods: Fasting glucose (mg/dl) at follow-up was compared between the mediterranean and usual groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a dietary intervention study on fasting glucose. The primary question concerns the Mediterranean diet intervention and fasting glucose (mg/dL) at follow-up. The accompanying dataset (data.csv) contains one row per participant-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

Methods

Participants and data. The export contains 437 records. Variables are listed in the data dictionary. Group membership is recorded in `diet`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only participants aged 18 to 80 years inclusive were analysed. Group labels in `diet` were entered free-text and were normalised by trimming whitespace and lower-casing before use. Records with missing glucose_fu were excluded (complete-case analysis).

Statistical analysis. Fasting glucose (mg/dl) at follow-up was compared between the mediterranean and usual groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). The U statistic is reported for the mediterranean group. The rank-biserial correlation was computed as $1 - 2U/(n_1 n_2)$. Medians and the difference in medians (mediterranean minus usual) are reported. All analyses used the cleaned analysis sample; no random resampling was used.

Results

The mediterranean group ($n = 202$, median 96.60) differed from the usual group ($n = 192$, median 105.35); the difference in medians was -8.75 ($U = 13059.00$, $p < 0.001$, rank-biserial $r = 0.327$).

Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.